

| SEQ NO | REV | NOTE NO. | ITEM NAME    | ITEM CODE | SIZE RANGE / SMALL END SIZES |      | DESCRIPTION                                                                                      | REFERENCE / LARGE END SIZES |     | NOTES |                                                                                                                                                                               |
|--------|-----|----------|--------------|-----------|------------------------------|------|--------------------------------------------------------------------------------------------------|-----------------------------|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10     |     |          |              |           |                              |      | ***** BALL VALVES *****                                                                          |                             |     | B1    | ALL BOLTS ARE METRIC DIA.                                                                                                                                                     |
| 20     |     |          |              |           |                              |      |                                                                                                  |                             |     | FF3   | USE FLAT FACE FLANGES WHEN MATING TO FRP FLANGES.                                                                                                                             |
| 20     |     |          | BALL         | BA1D1     | 0.75                         | 2    | BALL RB 150# RF ALLOY 20 BODY & TRIM 3 PIECE BOLTED BODY STANDARD PORT FILLED PTFE SEATS & SEALS | 0                           | 0   | FR6   | ALL FLANGE CONNECTIONS TO BE SUPPLIED WITH SPLASH GUARDS AROUND THE FLANGES.                                                                                                  |
| 30     |     |          | BALL         | BA1D2     | 0.75                         | 2    | BALL RB 2000# WOG SW ALLOY 20 BODY & TRIM 3 PIECE BOLTED BODY STANDARD PORT PTFE SEATS AND SEALS | 0                           | 0   | FS1   | USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.                                                                                               |
| 40     |     |          |              |           |                              |      | ***** PIPE *****                                                                                 |                             |     | FS9   | ONLY FOR MATCHING 300# FLANGED CONTROL VALVES.                                                                                                                                |
| 40     |     |          |              |           |                              |      |                                                                                                  |                             |     | SC1   | USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTROTRACE AND ORIFICE SETS |
| 50     |     |          | PIPE         | 5731001   | 0.5                          | 2    | PIPE SCH 40S SMLS ALLOY 20 B464 UNS N08020 BE                                                    | 0                           | 0   |       |                                                                                                                                                                               |
| 60     |     |          | PIPE         | 5731002   | 3                            | 4    | PIPE SCH 10S SMLS ALLOY 20 B464 UNS N08020 BE                                                    | 0                           | 0   |       |                                                                                                                                                                               |
| 70     |     |          |              |           |                              |      | ***** FITTINGS *****                                                                             |                             |     |       |                                                                                                                                                                               |
| 70     |     |          |              |           |                              |      |                                                                                                  |                             |     |       |                                                                                                                                                                               |
| 80     |     |          |              |           |                              |      |                                                                                                  |                             |     |       |                                                                                                                                                                               |
| 80     |     |          | ELBOW        | 5731011   | 0.5                          | 2    | ELL BW 90 DEG LR SCH 40S ALLOY 20 B366 UNS N08020                                                | 0                           | 0   |       |                                                                                                                                                                               |
| 90     |     |          | ELBOW        | 5731013   | 3                            | 4    | ELL 90 DEG LR SCH 10S ALLOY 20 B366 UNS N08020                                                   | 0                           | 0   | ZG3   | A STRAIGHT TEE WITH A CONCENTRIC REDUCER ON THE TEE BRANCH MAY BE USED AS A SUBSTITUTE FOR A REDUCING TEE.                                                                    |
| 100    |     |          |              |           |                              |      |                                                                                                  |                             |     |       |                                                                                                                                                                               |
| 100    |     |          | TEE          | 5464271   | 0.75                         | 2    | TEE BW SCH 40S ALLOY 20 B366 UNS N08020                                                          | 0                           | 0   |       |                                                                                                                                                                               |
| 110    |     |          | TEE          | 5731021   | 3                            | 4    | TEE BW SCH 10S ALLOY 20 B366 UNS N08020                                                          | 0                           | 0   | ZG4   | USE INSERT FOR REDUCING ON STRAIGHT SOCKET WELD TEE.                                                                                                                          |
| 120    |     | ZG3      | FLANGE       | 5731022   | 0.75                         | 1.5  | TEE BW RED SCH 40S ALLOY 20 B366 UNS N08020                                                      | 1                           | 2   |       |                                                                                                                                                                               |
| 130    |     |          | FLANGE       | 5731023   | 1.5                          | 2    | TEE BW RED SCH 10S/SCH 40S ALLOY 20 B366 UNS N08020                                              | 3                           | 4   |       |                                                                                                                                                                               |
| 140    |     |          | FLANGE       | 5731024   | 3                            | 3    | TEE BW RED SCH 10S ALLOY 20 B366 UNS N08020                                                      | 4                           | 4   |       |                                                                                                                                                                               |
| 150    |     |          |              |           |                              |      |                                                                                                  |                             |     |       |                                                                                                                                                                               |
| 150    |     |          | RED CONC     | 5731030   | 0.75                         | 1.5  | REDUCER CONC SCH 40S ALLOY 20 B366 UNS N08020                                                    | 1                           | 2   |       |                                                                                                                                                                               |
| 160    |     |          | RED CONC     | 5731031   | 1.5                          | 2    | REDUCER CONC SCH 10S/SCH 40S ALLOY 20 B366 UNS N08020                                            | 3                           | 4   |       |                                                                                                                                                                               |
| 170    |     |          | RED ECC      | 5464274   | 0.75                         | 1.5  | REDUCER ECC SCH 40S ALLOY 20 B366 UNS N08020                                                     | 1                           | 2   |       |                                                                                                                                                                               |
| 180    |     |          | RED ECC      | 5731040   | 1.5                          | 2    | REDUCER ECC SCH 10S/SCH 40S ALLOY 20 B366 UNS N08020                                             | 3                           | 4   |       |                                                                                                                                                                               |
| 190    |     |          |              |           |                              |      |                                                                                                  |                             |     |       |                                                                                                                                                                               |
| 190    |     |          | OLETS        | 5731050   | 0.5                          | 1    | WELDOLET SCH 40S ALLOY 20 B462 UNS N08020                                                        | 2                           | 2   |       |                                                                                                                                                                               |
| 200    |     |          | OLETS        | 5731051   | 0.5                          | 1.5  | WELDOLET SCH 10S/SCH 40S ALLOY 20 B462 UNS N08020                                                | 3                           | 4   |       |                                                                                                                                                                               |
| 210    |     |          |              |           |                              |      |                                                                                                  |                             |     |       |                                                                                                                                                                               |
| 210    |     |          | ELBOW        | 5731012   | 0.5                          | 2    | ELL 90 DEG 3000# SW ALLOY 20 B462 UNS N08020                                                     | 0                           | 0   |       |                                                                                                                                                                               |
| 220    |     |          |              |           |                              |      |                                                                                                  |                             |     |       |                                                                                                                                                                               |
| 220    |     | ZG4      | TEE          | 5731020   | 0.5                          | 2    | TEE 3000# SW ALLOY 20 B462 UNS N08020                                                            | 0                           | 0   |       |                                                                                                                                                                               |
| 230    |     |          |              |           |                              |      |                                                                                                  |                             |     |       |                                                                                                                                                                               |
| 230    |     |          |              |           |                              |      | ***** INSERTS *****                                                                              |                             |     |       |                                                                                                                                                                               |
| 240    |     |          |              |           |                              |      |                                                                                                  |                             |     |       |                                                                                                                                                                               |
| 240    |     | ZG4      | FITTING SCRD | 5731080   | 0.75                         | 1    | INSERT 3000# SW ALLOY 20 B462 UNS N08020                                                         | 1                           | 1.5 |       |                                                                                                                                                                               |
| 250    |     |          |              |           |                              |      |                                                                                                  |                             |     |       |                                                                                                                                                                               |
| 250    | U04 | FS1, FR6 |              |           |                              |      | ***** FLANGES *****                                                                              |                             |     |       |                                                                                                                                                                               |
| 260    |     |          |              |           |                              |      |                                                                                                  |                             |     |       |                                                                                                                                                                               |
| 260    |     | SC1      | FLANGE       | 5731067   | 0.75                         | 0.75 | FLG SCRD (FNPT) 150# RF ALLOY 20 B462 UNS N08020                                                 | 0                           | 0   |       |                                                                                                                                                                               |
| 270    |     |          | FLANGE       | 5731060   | 0.5                          | 2    | FLG SW 150# RF ALLOY 20 B462 UNS N08020 SCH 40S                                                  | 0                           | 0   |       |                                                                                                                                                                               |
| 280    |     | FF3      | FLANGE       | 5731063   | 0.5                          | 2    | FLG SW 150# FF ALLOY 20 B462 UNS N08020 SCH 40S                                                  | 0                           | 0   |       |                                                                                                                                                                               |
| 290    |     |          | FLANGE       | 5731066   | 0.5                          | 2    | FLG WN 150# RF ALLOY 20 B462 UNS N08020 SCH 40S                                                  | 0                           | 0   |       |                                                                                                                                                                               |
| 300    |     |          | FLANGE       | 5731069   | 3                            | 4    | FLG WN 150# RF ALLOY 20 B462 UNS N08020 SCH 10S                                                  | 0                           | 0   |       |                                                                                                                                                                               |
| 310    |     |          | BLIND        | 5731068   | 0.5                          | 4    | FLG BLIND 150# RF ALLOY 20 B462 UNS N08020                                                       | 0                           | 0   |       |                                                                                                                                                                               |
| 320    |     | FS9      | FLANGE       | 5731061   | 0.5                          | 2    | FLG WN 300# RF ALLOY 20 B462 UNS N08020 SCH 40S                                                  | 0                           | 0   |       |                                                                                                                                                                               |
| 330    |     |          | FLANGE       | 5731065   | 3                            | 4    | FLG WN 300# RF ALLOY 20 B462 UNS N08020 SCH 10S                                                  | 0                           | 0   |       |                                                                                                                                                                               |
| 340    |     |          |              |           |                              |      |                                                                                                  |                             |     |       |                                                                                                                                                                               |
| 340    |     |          |              |           |                              |      | ***** SWAGE NIPPLES *****                                                                        |                             |     |       |                                                                                                                                                                               |
| 350    |     |          |              |           |                              |      |                                                                                                  |                             |     |       |                                                                                                                                                                               |
| 350    |     |          | RED CONC     | 5721931   | 0.5                          | 0.5  | SWAGE CONC SCH 40S/80S B366 UNS N08020 PLE/TSE                                                   | 0.75                        | 1.5 |       |                                                                                                                                                                               |
| 360    |     |          | RED CONC     | 5721932   | 0.5                          | 0.5  | SWAGE CONC SCH 40S/80S B366 UNS N08020 BLE/TSE                                                   | 2                           | 2   |       |                                                                                                                                                                               |
| 370    |     |          |              |           |                              |      |                                                                                                  |                             |     |       |                                                                                                                                                                               |
| 370    |     | FS1      |              |           |                              |      | ***** GASKETS *****                                                                              |                             |     |       |                                                                                                                                                                               |
| 380    |     |          |              |           |                              |      |                                                                                                  |                             |     |       |                                                                                                                                                                               |
| 380    |     |          | GASKET       | 5673043   | 0.75                         | 4    | GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR                                                       | 0                           | 0   |       |                                                                                                                                                                               |
| 390    |     |          | GASKET       | 5731070   | 0.5                          | 4    | GASKET 150# LOW STRESS TEFLON ENVELOPE NEOPRENE FILLER, GARLOCK MODEL 370                        | 0                           | 0   |       |                                                                                                                                                                               |
| 400    |     | FS9      | GASKET       | 5731071   | 0.5                          | 4    | GASKET 300# LOW STRESS TEFLON ENVELOPE NEOPRENE FILLER, GARLOCK MODEL 370                        | 0                           | 0   |       |                                                                                                                                                                               |
| 410    |     |          |              |           |                              |      |                                                                                                  |                             |     |       |                                                                                                                                                                               |
| 410    |     |          |              |           |                              |      | ***** BOLTING *****                                                                              |                             |     |       |                                                                                                                                                                               |
| 420    |     |          |              |           |                              |      |                                                                                                  |                             |     |       |                                                                                                                                                                               |
| 420    |     | B1       | BOLTING      | 5675770   | 0.5                          | 99   | STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS                                                | 0                           | 0   |       |                                                                                                                                                                               |
| 430    |     |          |              |           |                              |      |                                                                                                  |                             |     |       |                                                                                                                                                                               |

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|------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------|------------|-----|----------------|-------------------|
|  | TENGIZCHEVROI<br>TCO<br>TENGIZ,<br>KAZAKHSTAN | CONSTRUCTION                                                                                                         | CORR.ALLOW.       | SERVICE                                                                      | MATERIAL                                                                                                                              | DATE       | BY         | REV | SHEET          | PIPING LINE CLASS |
|                                                                                    |                                               | ALL SIZES - BUTT WELD<br>SEAL WELDING - FOR THREADED CONNECTIONS<br>PIPE BENDS - NONE<br>THREAD SEALANT - PTFE PASTE | CORR. ALLOW. 0 MM | DESIGN ANSI B16.5 MG 3.1<br>CODE ASME B31.3<br>FLUID CODES - PIM-SU-5112-TCO | MATERIAL: ALLOY 20<br>FLANGE RATING: 150 RF<br>TEMPERATURE RANGE: -40 / 200 °C<br>COMMENT: 1/2" & BELOW -<br>FOR VENDOR PACKAGES ONLY | 12/16/2024 | BY<br>OZAC | U04 | Page 1<br>of 1 | 150A20            |